Healthcare Analytics and Machine Learning

Official Presentation Script

Genie Agent, Gold-Layer Star Schema, Logistic Regression, Model Evaluation, MLflow and UAT

End-to-end story: Trusted healthcare data -> Natural-language analytics -> Predictive machine learning -> Controlled validation

Prepared for: Project presentation / SME discussion

Recommended speaking time: Approximately 7-9 minutes

Presentation approach: Explain each theory term at the point where it appears in the project flow.

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Presentation Roadmap

Gold Layer	**Star Schema**	**Genie Agent**
Natural-language Analytics	**ML Objective**	**Preprocessing**
Train-Test Split	**Model Training**	**Evaluation Metrics**
Model Selection	**MLflow**	**UAT**

Core transition: Genie helps users understand existing data. Machine learning uses patterns in the same trusted data to predict an outcome.

1. Genie Agent and Gold-Layer Star Schema (45-60 seconds)

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Before moving into machine learning, I will briefly explain how we used Genie with our Gold-layer data.

After completing the Bronze, Silver and Gold layers, we have clean, business-ready healthcare data in the Gold layer. We organized this data using a star schema, where a central fact table is connected with related dimension tables.

Once the star schema was ready, we created a Genie agent and provided the relevant Gold-layer tables as its data context. The main purpose of Genie is to provide self-service analytics through natural language. Instead of writing SQL queries or manually understanding table joins, a business user can ask a question such as, “How many patients are present in each age band?”

Genie uses the configured business data and context to return an analytical result. In simple terms, Genie acts as a bridge between business users and our trusted Gold-layer data.

Until this point, we are mainly using data for analysis. Next, we take the same trusted data further and use its patterns for prediction. This brings us to the machine-learning flow.

Gold Layer -> Star Schema -> Genie Agent -> Natural-language Questions -> Business Insights -> ML Prediction

2. Machine-Learning Objective (35 seconds)

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The first step in our machine-learning flow is defining what we want to predict. In our case, the target is patient status, which contains three categories: ACTIVE, INACTIVE and PENDING.

Because we are predicting a category with more than two possible classes, this is a multiclass classification problem. The input features are age band, sex, city, insurance type and primary language, while patient status is the target variable.

Features are the input information given to the model. The target is the output that we want the model to predict.

Theory terms

- Classification: Predicting a category rather than a continuous numerical value.
- Multiclass classification: Classification with more than two possible target classes.
- Feature: An input variable used by the model.
- Target: The outcome the model is trained to predict.

3. Data Preprocessing (50 seconds)

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Once we select the features and target, the next step is data preprocessing. Most of our selected features are categorical, so they must be converted into a numerical form the models can process.

First, missing categorical values are handled using imputation. In our implementation, missing values are replaced using the most frequently occurring category. Then we use One-Hot Encoding, which converts each category into numerical indicator columns.

These transformations are included in a pipeline so the same preparation is applied consistently during training and prediction.

Preprocessing = Missing-value handling + Categorical encoding + Consistent transformation pipeline

Key theory

- Imputation: Replacing missing values using a defined method.
- One-Hot Encoding: Representing each category through separate numerical indicator columns.
- Pipeline: A controlled sequence of preprocessing and model-training steps.

4. Train-Test Split and Class Balance (50 seconds)

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After preprocessing, we divide the data into training and testing datasets using an 80-20 split. Approximately 80 percent is used for training, and 20 percent is reserved for testing.

Training data teaches the model the patterns. Testing data evaluates the model on records it did not see during training. We also use stratification to preserve the distribution of ACTIVE, INACTIVE and PENDING classes across both datasets.

We also consider class imbalance. If one class contains far more records than another, the model can become biased toward the majority class. Balanced class weights help give appropriate importance to each class during training.

Training data teaches the model. Testing data evaluates the model.

Why this matters

- Testing on unseen records gives a more realistic estimate of model generalization.
- Stratification reduces the risk of an unrepresentative train-test split.
- Balanced class weights help reduce majority-class bias.

5. Model Training: Logistic Regression and Random Forest (1 minute)

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Now that the data is prepared, we train two classification models: Logistic Regression and Random Forest.

Even though Logistic Regression contains the word “regression,” it is commonly used for classification. It learns relationships between the input features and target classes, and uses those relationships to classify new records.

In standard binary Logistic Regression, the sigmoid function converts a raw model score into a probability between zero and one. Our project contains three target classes, so it is a multiclass classification use case rather than a simple positive-versus-negative problem.

Random Forest is an ensemble-learning algorithm. It combines the predictions of multiple decision trees. Training both algorithms allows us to compare their performance on the same held-out test data.

Logistic Regression Interpretable classification baseline; models class relationships.	Random Forest Combines multiple decision trees to capture more complex patterns.

6. Confusion Matrix (55 seconds)

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After training, we generate predictions and evaluate them. The Confusion Matrix compares the actual class with the class predicted by the model.

In binary classification, a True Positive is correctly predicted positive, a True Negative is correctly predicted negative, a False Positive is incorrectly predicted positive, and a False Negative is incorrectly predicted negative.

Our problem is multiclass. Therefore, the matrix compares ACTIVE, INACTIVE and PENDING. The diagonal cells represent correct classifications, while off-diagonal cells show which patient-status classes the model confuses with one another.

Term	Meaning
True Positive	Positive class correctly identified
True Negative	Negative class correctly identified
False Positive	Predicted positive, but actually negative
False Negative	Predicted negative, but actually positive

7. Model Evaluation Metrics (1 minute 30 seconds)

Metric	Formula	Interpretation	Logistic Regression
Accuracy	Correct predictions / Total predictions	How many predictions were correct overall?	40.54%
Precision	TP / (TP + FP)	When the model predicts a class, how often is it correct?	38.51% weighted
Recall	TP / (TP + FN)	Out of the actual cases, how many did the model identify?	40.54% weighted
F1 Score	2 x (Precision x Recall) / (Precision + Recall)	How well are precision and recall balanced?	39.02% weighted

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Accuracy gives the overall percentage of correct predictions. However, accuracy alone may hide the types of mistakes being made.

Precision measures the reliability of a predicted class. Recall measures how much of an actual class the model successfully identifies. The F1 Score is the harmonic mean of precision and recall, so it provides one measure that balances both.

For multiclass evaluation, our notebook reports weighted precision, recall and F1, which summarize class-level results while considering the number of records in each class.

8. Model Comparison and Selection (40 seconds)

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After calculating the metrics, we compare both models. Logistic Regression achieved approximately 40.54 percent test accuracy, while Random Forest achieved approximately 35.14 percent.

Therefore, according to the notebook’s selection logic, Logistic Regression is selected as the better-performing model of the two.

However, the selected result remains below the notebook’s expected accuracy threshold of 70 percent. This means the current result should be treated as a baseline that requires further improvement, not as a highly accurate production model.

Model	Test Accuracy	Decision
Logistic Regression	40.54%	Selected as the better of the two
Random Forest	35.14%	Not selected

Official wording: “Logistic Regression performed better than Random Forest in the current comparison, but the result remains below the desired threshold.”

9. MLflow and UAT (55 seconds)

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After selecting the model, we use MLflow to track the machine-learning lifecycle. MLflow records model parameters, evaluation metrics and trained model artifacts. This provides experiment traceability and reproducibility.

Finally, we perform User Acceptance Testing, or UAT, for the broader healthcare data solution. UAT and ML evaluation validate different areas. Accuracy, Precision, Recall, F1 Score and the Confusion Matrix measure predictive performance. UAT checks whether the underlying data product satisfies expected business and data-quality conditions.

Our UAT includes checks such as Silver-versus-Gold patient counts, encounter counts, age-band completeness and key-quality validation. The captured run completed with a SUCCESS status.

ML Evaluation	UAT
Checks prediction performance using the confusion matrix and classification metrics.	Checks the broader data product against business and data-quality expectations.

10. Final Closing (30 seconds)

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To summarize, our complete flow starts with trusted Gold-layer healthcare data organized using a star schema. We make this data accessible to business users through Genie and natural-language analytics.

We then move from analytics to prediction. We identify our features and target, preprocess the data, create the train-test split, handle class imbalance, and train Logistic Regression and Random Forest.

We evaluate both models using the Confusion Matrix, Accuracy, Precision, Recall and F1 Score. Logistic Regression performs better in the current comparison, so it is selected and tracked through MLflow. Finally, UAT validates the broader healthcare data solution.

Overall, the project takes us from trusted data, to business analytics, to predictive machine learning, and finally to controlled validation.

Bronze -> Silver -> Gold -> Star Schema -> Genie -> Analytics -> ML -> Evaluation -> MLflow -> UAT

End of presentation script

Quick Reference: Terms to Remember

Term	Presentation-ready meaning
Star schema	A central fact table connected to descriptive dimension tables.
Genie agent	A natural-language interface for exploring governed business data.
Self-service analytics	Business users explore data without manually writing every SQL query.
Multiclass classification	Predicting one class from more than two possible categories.
One-Hot Encoding	Converting categorical values into numerical indicator columns.
Stratification	Maintaining class distribution across training and testing sets.
Class imbalance	Unequal record distribution across target classes.
Logistic Regression	A classification algorithm that models class relationships and probabilities.
Random Forest	An ensemble model combining multiple decision trees.
Confusion Matrix	A comparison of actual and predicted classes.
Accuracy	Overall percentage of correct predictions.
Precision	Reliability of predicted classes.
Recall	Proportion of actual cases successfully identified.
F1 Score	Harmonic mean balancing precision and recall.
MLflow	Experiment and model lifecycle tracking.
UAT	Validation against business and data-quality expectations.